KAIZEN MATHS

Lesson starter

Volumes of Revolution

Disc Method

Name _____

S1

Use the disc method to find each volume.

The region under $y = 2x + 1$ from $x = 1$ to $x = 3$ is rotated about the x-axis. Find the volume.

S2

Use the disc method to find each volume.

The region under $y = x^1$ from $x = 0$ to $x = 3$ is rotated about the x-axis. Find the volume.

S3

Square the square-root radius, then integrate.

The region under $y = \sqrt{x + 3}$ from $x = 0$ to $x = 4$ is rotated about the x-axis. Find the volume.

S4

Use the disc method to find each volume.

The region under $y = 3x + 2$ from $x = 1$ to $x = 4$ is rotated about the x-axis. Find the volume.

Show your working

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Use the disc method to find each volume.

The region under $y = x^1$ from $x = 0$ to $x = 3$ is rotated about the x-axis. Find the volume.

S3

Square the square-root radius, then integrate.

The region under $y = \sqrt{x + 3}$ from $x = 0$ to $x = 4$ is rotated about the x-axis. Find the volume.

S4

Use the disc method to find each volume.

The region under $y = 3x + 2$ from $x = 1$ to $x = 4$ is rotated about the x-axis. Find the volume.

ALG-3EBB-L1 · S

Show your working

KAIZEN MATHS

Exit ticket

Volumes of Revolution

Disc Method

Name

E1

Use the disc method to find each volume.

The region under $y = 3x^2$ from $x = 0$ to $x = 2$ is rotated about the x-axis. Find the volume.

E2

Square the square-root radius, then integrate.

The region under $y = \sqrt{x + 2}$ from $x = 0$ to $x = 3$ is rotated about the x-axis. Find the volume.

E3

Use the disc method to find each volume.

The region under $y = 2x + 2$ from $x = 0$ to $x = 2$ is rotated about the x-axis. Find the volume.

KAIZEN MATHS

Exit ticket

Volumes of Revolution

Disc Method

Name

E1

Use the disc method to find each volume.

The region under $y = 3x^2$ from $x = 0$ to $x = 2$ is rotated about the x-axis. Find the volume.

E2

Square the square-root radius, then integrate.

The region under $y = \sqrt{x + 2}$ from $x = 0$ to $x = 3$ is rotated about the x-axis. Find the volume.

E3

Use the disc method to find each volume.

The region under $y = 2x + 2$ from $x = 0$ to $x = 2$ is rotated about the x-axis. Find the volume.

ALG-3EBB-L1 · E

Show your working

KAIZEN MATHS

Exit ticket

Volumes of Revolution

Disc Method

Name

E1

Use the disc method to find each volume.

The region under $y = 3x^2$ from $x = 0$ to $x = 2$ is rotated about the x-axis. Find the volume.

E2

Square the square-root radius, then integrate.

The region under $y = \sqrt{x + 2}$ from $x = 0$ to $x = 3$ is rotated about the x-axis. Find the volume.

E3

Use the disc method to find each volume.

The region under $y = 2x + 2$ from $x = 0$ to $x = 2$ is rotated about the x-axis. Find the volume.

ALG-3EBB-L1 · E

Show your working

KAIZEN MATHS

Exit ticket

Volumes of Revolution

Disc Method

Name

E1

Use the disc method to find each volume.

The region under $y = 3x^2$ from $x = 0$ to $x = 2$ is rotated about the x-axis. Find the volume.

E2

Square the square-root radius, then integrate.

The region under $y = \sqrt{x + 2}$ from $x = 0$ to $x = 3$ is rotated about the x-axis. Find the volume.

E3

Use the disc method to find each volume.

The region under $y = 2x + 2$ from $x = 0$ to $x = 2$ is rotated about the x-axis. Find the volume.

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Show your working

Teacher answers & working

Volumes of Revolution · Disc Method

ALG-3EBB-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Use the disc method to find each volume.

The region under $y = 2x + 1$ from $x = 1$ to $x = 3$ is rotated about the x-axis. Find the volume.

Answer:

$V = \dfrac{158\pi}{3}$

Working:

For rotation about the x-axis, use $V = \pi \int_a^b y^2 \, dx$.

Here $y = 2x + 1$, $a = 1$ and $b = 3$, so $V = \pi \int_1^3 (2x + 1)^2 \, dx$.

Square the radius: $(2x + 1)^2 = 4x^2 + 4x + 1$.

So $V = \pi \int_1^3 (4x^2 + 4x + 1) \, dx$.

Integrate: $\int (4x^2 + 4x + 1) \, dx = \dfrac{4}{3}x^3 + 2x^2 + x$.

Substitute the bounds: $\left[\dfrac{4}{3}x^3 + 2x^2 + x\right]_1^3 = \dfrac{158}{3}$.

Multiply by π: $V = \dfrac{158\pi}{3}$.

S2

Use the disc method to find each volume.

The region under $y = x^1$ from $x = 0$ to $x = 3$ is rotated about the x-axis. Find the volume.

Answer:

$V = 9\pi$

Working:

For rotation about the x-axis, use $V = \pi \int_a^b y^2 \, dx$.

Here $y = x^1$ and the bounds are 0 to 3.

Square the radius: $(x^1)^2 = x^2$.

Set up the volume: $V = \pi \int_0^3 x^2 \, dx$.

Integrate: $\int x^2 \, dx = \dfrac{1}{3}x^3$.

Substitute the bounds: $\left[\dfrac{1}{3}x^3\right]_0^3 = 9$.

Multiply by π: $V = 9\pi$.

Teacher answers & working

Volumes of Revolution · Disc Method

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S3

Square the square-root radius, then integrate.

The region under $y = \sqrt{x+3}$ from $x = 0$ to $x = 4$ is rotated about the x-axis. Find the volume.

Answer:

$V = 20\pi$

Working:

For rotation about the x-axis, use $V = \pi \int_a^b y^2 \, dx$.

Here $y = \sqrt{x+3}$, so $y^2 = x + 3$.

Set up the volume: $V = \pi \int_0^4 (x+3) \, dx$.

Integrate: $\int (x+3) \, dx = \frac{1}{2}x^2 + 3x$.

Substitute the bounds: $\left[\frac{1}{2}x^2 + 3x\right]_0^4 = 20$.

Multiply by π: $V = 20\pi$.

S4

Use the disc method to find each volume.

The region under $y = 3x + 2$ from $x = 1$ to $x = 4$ is rotated about the x-axis. Find the volume.

Answer:

$V = 291\pi$

Working:

For rotation about the x-axis, use $V = \pi \int_a^b y^2 \, dx$.

Here $y = 3x + 2$, $a = 1$ and $b = 4$, so $V = \pi \int_1^4 (3x+2)^2 \, dx$.

Square the radius: $(3x+2)^2 = 9x^2 + 12x + 4$.

So $V = \pi \int_1^4 (9x^2 + 12x + 4) \, dx$.

Integrate: $\int (9x^2 + 12x + 4) \, dx = 3x^3 + 6x^2 + 4x$.

Substitute the bounds: $\left[3x^3 + 6x^2 + 4x\right]_1^4 = 291$.

Multiply by π: $V = 291\pi$.

Teacher answers & working

Volumes of Revolution · Disc Method

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E1

Use the disc method to find each volume.

The region under $y = 3x^2$ from $x = 0$ to $x = 2$ is rotated about the x-axis. Find the volume.

Answer:

$V = \dfrac{288\pi}{5}$

Working:

For rotation about the x-axis, use $V = \pi \int_a^b y^2 \, dx$.

Here $y = 3x^2$ and the bounds are 0 to 2.

Square the radius: $(3x^2)^2 = 9x^4$.

Set up the volume: $V = \pi \int_0^2 9x^4 \, dx$.

Integrate: $\int 9x^4 \, dx = \dfrac{9}{5}x^5$.

Substitute the bounds: $\left[\dfrac{9}{5}x^5\right]_0^2 = \dfrac{288}{5}$.

Multiply by π: $V = \dfrac{288\pi}{5}$.

E2

Square the square-root radius, then integrate.

The region under $y = \sqrt{x + 2}$ from $x = 0$ to $x = 3$ is rotated about the x-axis. Find the volume.

Answer:

$V = \dfrac{21\pi}{2}$

Working:

For rotation about the x-axis, use $V = \pi \int_a^b y^2 \, dx$.

Here $y = \sqrt{x + 2}$, so $y^2 = x + 2$.

Set up the volume: $V = \pi \int_0^3 (x + 2) \, dx$.

Integrate: $\int (x + 2) \, dx = \dfrac{1}{2}x^2 + 2x$.

Substitute the bounds: $\left[\dfrac{1}{2}x^2 + 2x\right]_0^3 = \dfrac{21}{2}$.

Multiply by π: $V = \dfrac{21\pi}{2}$.

Teacher answers & working

Volumes of Revolution · Disc Method

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E3

Use the disc method to find each volume.

The region under $y = 2x + 2$ from $x = 0$ to $x = 2$ is rotated about the x-axis. Find the volume.

Answer:

$V = \dfrac{104\pi}{3}$

Working:

For rotation about the x-axis, use $V = \pi \int_a^b y^2 \, dx$.

Here $y = 2x + 2$, $a = 0$ and $b = 2$, so $V = \pi \int_0^2 (2x + 2)^2 \, dx$.

Square the radius: $(2x + 2)^2 = 4x^2 + 8x + 4$.

So $V = \pi \int_0^2 (4x^2 + 8x + 4) \, dx$.

Integrate: $\int (4x^2 + 8x + 4) \, dx = \dfrac{4}{3}x^3 + 4x^2 + 4x$.

Substitute the bounds: $\left[\dfrac{4}{3}x^3 + 4x^2 + 4x\right]_0^2 = \dfrac{104}{3}$.

Multiply by π: $V = \dfrac{104\pi}{3}$.